

MOBRA: A Requirements-Based Operational BioSecurity Readiness Framework for Mobile Biological Laboratories

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Highlights

- A requirements-based operational biosecurity readiness framework was developed for mobile biological laboratories.
- WHO, ISO and BMBL guidance was translated into measurable operational requirements.
- MOBRA links objective evidence, representative hazards, risk prioritization, and readiness assessment within one traceable framework.
- A publicly accessible Streamlit implementation supports dataset ingestion, readiness indicators, hazard heat maps, interactive decision-support dashboards, and reproducible exports.
- The framework is ready for expert validation, reliability assessment, and multi-site operational evaluation.
- Aim :To develop a requirements-based operational biosecurity readiness framework and define its validation strategy.

Abstract

Mobile biological laboratories extend diagnostic capacity to outbreak sites and resource-constrained settings, but mobility creates unique operational challenges, including variable infrastructure, logistics, information security, specimen accountability, and containment risks. This study presents MOBRA, a requirements-based operational biosecurity readiness framework, together with a transparent Python prototype that automates readiness assessment, risk visualization, and decision-support analytics. A design science approach integrated the WHO Laboratory Biosafety Manual, WHO Laboratory Biosecurity Guidance, WHO Minimum Operational Standards for Rapid Response Mobile Laboratories, ISO 35001, ISO 31000, the sixth edition of Biosafety in Microbiological and Biomedical Laboratories (BMBL), recent scientific literature, the author's original Jordanian biosecurity proposal, and the technical characteristics of a Jordanian truck-based mobile biological laboratory.

The framework comprises seven integrated components: an initial Operational Requirements Library (60 requirements across ten domains), an evidence-based 0–5 assessment scale, a representative hazard register (**24 hazards**), a **5 × 5 likelihood–consequence risk matrix**, **color-coded risk classification**, **Quantitative domain-readiness indicators**, and an **overall Biosecurity Readiness Index (BRI)**. The Python prototype performs automated validation of input structure and score consistency, calculates risk and readiness, and generates color-coded hazard heat maps together with interactive decision-support dashboards.

A synthetic demonstration dataset produced an illustrative overall BRI of 81.0%. The software was subsequently extended as a modular Streamlit dashboard capable of importing CSV, Excel, and JSON-compatible data; assessing data quality; mapping heterogeneous columns; calculating risk and readiness outputs; and exporting analyzed datasets and reports. Three external-data categories were identified for computational verification: Canadian laboratory incident reports, Global Health Security Index 2021 indicators, and high-risk pathogen exposure-event records. These sources are used to test software interoperability and analytical behavior only and do not constitute scientific validation of MOBRA. The live dashboard and source code are publicly available. All prototype outputs remain illustrative and must not be interpreted as institutional performance measurements or operational assessments. Formal expert content validation, inter-assessor reliability testing, software validation, and multi-site operational evaluation are required before institutional adoption.

Keywords: mobile biological laboratory; biosecurity; biorisk management; operational readiness; ISO 35001; risk matrix; heat map; Python.

1. Introduction

Rapid response mobile biological laboratories provide rapidly deployable diagnostic capacity during infectious disease outbreaks, public health emergencies, and other field operations. The World Health Organization (WHO) Minimum Operational Standards for Rapid Response Mobile Laboratories (2025) describe these laboratories as coordinated, interoperable, quality-assured, safe, scalable, and predictable assets. Mobility, however, fundamentally alters the operational risk environment. Biological materials and equipment must be transported safely; utilities may be temporary, access-control arrangements vary between deployment sites, specimen traceability and chain-of-custody requirements become more complex, information systems may be disrupted, and personnel frequently operate under significant time pressure.

The WHO Laboratory Biosafety Manual (4th edition) places activity-specific risk assessment at the center of proportionate biosafety. The WHO Laboratory Biosecurity Guidance (2024) complements this approach by addressing the protection of high-consequence biological materials, technology, and information throughout their life cycle. ISO 35001 defines a management system for identifying, assessing, controlling, and monitoring biorisks, while ISO 31000 provides a structured framework of risk management principles. The sixth edition of Biosafety in Microbiological and Biomedical Laboratories (BMBL) remains a principal technical reference for laboratory practices and containment. Collectively, these international references establish a robust scientific and regulatory foundation for biosafety, biosecurity, and biorisk management. However, they primarily provide guidance rather than an integrated operational assessment framework specifically designed for mobile biological laboratories.

Despite this strong international foundation, laboratory managers and field decision-makers still require a practical mechanism for translating international guidance into measurable operational readiness decisions. Conventional checklists may identify missing controls. However, they rarely provide a transparent and traceable framework linking operational requirements, objective evidence, representative hazards, quantified risk, and corrective actions within a unified operational assessment process. The conceptual basis of the original Jordanian biosecurity proposal similarly identified unauthorized access, biological material diversion, protection of sensitive information, and inconsistent procedural compliance as key operational challenges requiring an integrated assessment approach.

The operational reference environment for MOBRA was a truck-based, negative-pressure mobile biological laboratory designed for field deployment under high-containment conditions. The platform was equipped with engineering containment systems, including a Class III glovebox and two Class II biological safety cabinets, together with molecular diagnostic instruments, automated nucleic-acid extraction systems, sterilization and decontamination equipment, cold storage, and supporting analytical infrastructure. Its modular design allows compatible analytical instruments to be added according to mission requirements. This operational environment demonstrates why engineering controls, specimen accountability, personnel competency, information protection, and logistical continuity must be evaluated as an integrated operational system rather than as isolated components. The platform served as the operational reference model for the initial design and development of the MOBRA framework.

Accordingly, MOBRA was developed to translate international guidance into a transparent, evidence-based operational decision-support framework for mobile biological laboratories.

This study aimed to design, develop, and implement MOBRA as a transparent, requirements-based operational biosecurity readiness and decision-support framework, supported by a transparent and reproducible Python prototype. The study presents the development and proof-of-concept implementation of the framework and its accompanying prototype. Formal expert validation, inter-assessor reliability testing, standardized acceptance thresholds, and institutional performance assessment remain beyond the scope of the present study.

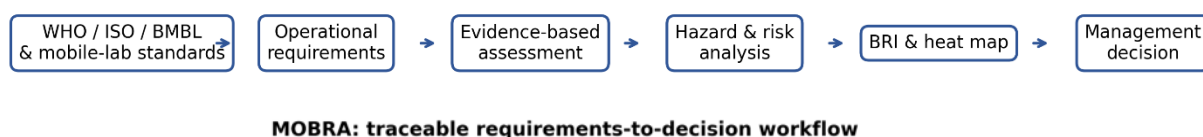


Figure 1. MOBRA graphical abstract and traceability chain. International guidance is converted into operational requirements, evidence, risk prioritization, readiness indicators, and management action.

Key innovations of MOBRA
• Requirements-based operational assessment across ten domains.
• Evidence-based six-level scoring rather than binary compliance checks.
• A representative hazard registered linked to likelihood-consequence risk mapping.
• An aggregate Biosecurity Readiness Index interpreted with critical-risk overrides.
• A transparent Python prototype supporting reproducible analytics and decision support.

2. Materials and methods

2.1 Study design and evidence base

A **Design Science Research (DSR)** approach was selected because the primary outcome of this study was the development of an operational biosecurity assessment framework and its supporting software prototype. The methodology proposed by Peffers et al. guided problem identification, definition of solution objectives, framework design, prototype development, demonstration, and evaluation planning.

The normative evidence base comprised **World Health Organization (WHO)** guidance on laboratory biosafety, laboratory biosecurity, rapid response mobile laboratories, and the transport of infectious substances; **ISO 35001** for biorisk management; **ISO 31000** for risk management; and the sixth edition of *Biosafety in Microbiological and Biomedical Laboratories (BMBL)*. Supporting evidence included recent scientific literature on mobile biological laboratories and laboratory biosafety and biosecurity, established content-validation methodologies, the author's original Jordanian biosecurity proposal, and the technical documentation of the reference mobile biological laboratory. Collectively, these sources informed the identification of operational requirements, contextual adaptation of the framework, prototype development, and the design of the proposed validation strategy.

2.2 Operational case context

The operational reference environment was a self-contained, truck-based biological laboratory operated by the Jordan Armed Forces–Royal Medical Services (**JAF-RMS**). The approximately **42 m²** platform operates under **negative pressure** relative to the entry area. Potentially hazardous specimens are transferred through a **pass box into a Class III glovebox**, minimizing specimen movement through the entry zone.

The laboratory is equipped with two Class II biological safety cabinets, real-time PCR systems, automated nucleic-acid extraction systems, centrifuges, an autoclave, ELISA instruments, spectrophotometric capability, refrigeration, and supporting decontamination infrastructure. Its modular design allows compatible analytical instruments to be added according to specific mission and diagnostic requirements.

The laboratory was used solely to define the operational context and inform the initial design of the MOBRA framework. It served as a reference operational environment rather than as the subject of institutional performance evaluation. No confidential operational records, patient-identifiable information, or institutional performance data were accessed or used during framework development or prototype demonstration.

2.3 Operational Requirements Library (ORL)

The initial Operational Requirements Library (ORL) comprises 60 operational requirements distributed across ten assessment domains, with six requirements initially allocated to each domain. The ORL constitutes the core knowledge base of the MOBRA framework and provides the primary link between international guidance and measurable operational assessment.

Each requirement record includes a unique identifier, an operational requirement statement, expected objective evidence, an associated representative hazard, and a maximum assessment score. This structure enables individual requirements to be traced to the evidence required for verification and to the operational risks they are intended to address.

The balanced distribution of requirements across the ten domains was adopted as an initial prototype design decision to provide comparable domain coverage and practical usability. The number, distribution, and relative importance of requirements may be refined following expert content validation, reliability assessment, and operational field testing.

Requirements were drafted as measurable operational expectations rather than direct reproductions of copyrighted standard clauses. Formal clause-level mapping to individual WHO, ISO, and BMBL documents should be undertaken only by authorized institutional reviewers with lawful access to the relevant standards.

Domain	Initial requirements
Emergency Incident Management	6
Engineering Controls	6
Information Security	6
Laboratory Operations	6
Mission Planning	6
Personnel Reliability	6
Physical Security	6
Specimen Security	6
Transportation Logistics	6
Waste Management Continuous Improvement	6

Table 1. Distribution of the initial 60 operational requirements across the ten MOBRA assessment domains

2.4 Evidence-based scoring

Each operational requirement is assessed using a **six-level maturity scale ranging from 0 to 5**. Scores are assigned as follows: **0**, not implemented; **1**, informal or undocumented; **2**, partially implemented; **3**, implemented; **4**, implemented and objectively verified; and **5**, verified and routinely reviewed or continuously improved. The six-level scale was selected to distinguish progressive stages of implementation, objective verification, and continuous improvement while maintaining practical applicability during operational assessments.

Objective evidence may include approved procedures, direct observation, access-control and chain-of-custody records, equipment certification and calibration records, training and competency documentation, electronic audit trails, incident reports, and corrective and preventive action (CAPA) records.

Domain readiness is calculated as the percentage of the total observed scores relative to the maximum achievable score for all requirements within the corresponding domain:

$$\text{Domain Readiness (\%)} = \frac{\sum \text{Observed Scores}}{\sum \text{Maximum Scores}} \times 100$$

This approach provides a quantitative domain-level readiness indicator while preserving traceability to the individual requirement and the objective evidence supporting each assigned score.

Interpretation	Score
Not implemented	0
Informal or undocumented	1
Partially implemented	2
Implemented	3
Implemented and verified	4
Verified and routinely reviewed or continuously improved	5

Table 2. MOBRA six-level evidence-based assessment scale.

Note: Scores represent progressive levels of implementation maturity and evidence strength. Final scoring criteria and assessor guidance require formal expert validation before operational application.

2.5 Hazard and risk Assessment

Representative hazards were linked to operational requirements to ensure that each requirement could be interpreted within its corresponding operational risk context. Hazards were intentionally distributed across the ten MOBRA assessment domains to provide representative coverage of operational biosecurity risks associated with mobile biological laboratory deployment.

Risk was calculated as the product of **likelihood (L)** and **consequence (C)**, with each parameter scored on a **five-point scale ranging from 1 to 5**:

$$\text{Risk Score} = \text{Likelihood} \times \text{Consequence}$$

Prototype risk categories classified scores of **1–4 as low**, **5–9 as moderate**, **10–16 as high**, and **17–25 as extreme**. These prototype risk categories represent framework design rules rather than internationally standardized risk-acceptance thresholds and should therefore be aligned with institutional risk appetite, governance requirements, and escalation procedures during formal validation and operational implementation.

Accordingly, deployment decisions should consider both aggregate readiness and critical-risk status. The framework incorporates a critical-control override, whereby an extreme residual risk or failure of a mission-critical containment or biosecurity control may prevent deployment approval regardless of the overall **Biosecurity Readiness Index (BRI)**.

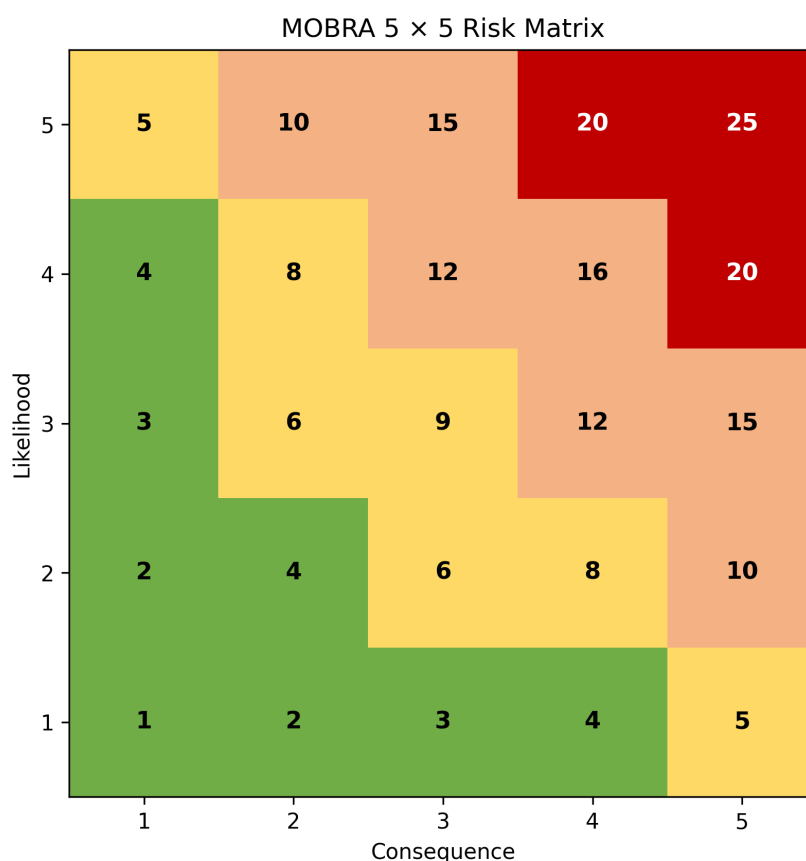


Figure 2. MOBRA 5 × 5 likelihood–consequence risk matrix. Risk score is calculated as likelihood × consequence. Prototype risk categories are defined as low (Green, 1–4), moderate (Yellow, 5–9), high (Orange, 10–16), and extreme (Red, 17–25). These categories are framework design rules and require institutional validation to define context-specific risk-acceptance and escalation criteria.

2.6 Biosecurity Readiness Index

The overall **Biosecurity Readiness Index (BRI)** provides an aggregate quantitative summary of operational readiness across all MOBRA assessment domains. In the initial unweighted prototype, the BRI is calculated as the percentage of the total observed requirement scores relative to the maximum achievable score across all assessed operational

$$\text{BRI (\%)} = \left(\frac{\sum \text{Observed Requirement Scores}}{\sum \text{Maximum Requirement Scores}} \right) \times 100$$

requirements:

Equal weighting was adopted during initial framework development to preserve transparency, reproducibility, and ease of interpretation. Future application of criticality-based or expert-derived weighting should be supported by formal content validation and sensitivity analysis to determine whether alternative weighting approaches improve the validity and operational usefulness of the index.

The BRI is intended as a **decision-support indicator rather than a standalone deployment criterion**. Accordingly, the overall index should be interpreted together with domain-level readiness, the representative hazard register, the quality and completeness of objective evidence, residual risk status, and any critical-control findings. A high aggregate BRI, therefore, does not override an extreme residual risk or the failure of a mission-critical containment or biosecurity control.

2.7 Python prototype

Implementation of the MOBRA framework was demonstrated through a modular Python 3.11 application deployed with Streamlit. Pandas and NumPy support data processing, Plotly provides interactive visualizations, OpenPyXL supports spreadsheet export, and Jinja2 supports reproducible HTML reporting. The application is designed to run its core analytical functions without requiring external internet services.

The application accepts CSV, XLSX, XLS, and, where structurally compatible, JSON files. It previews imported records, identifies data types, missing values and duplicate rows, supports user-guided column mapping, standardizes heterogeneous field names, classifies incident records, validates likelihood and consequence scores, calculates risk categories, generates a 5 × 5 risk matrix, and calculates the Biosecurity Readiness Index when requirement-level assessment data are available. Outputs include KPI cards, Plotly charts, summary tables, data-quality diagnostics, HTML reports, cleaned and analyzed CSV files, and multi-sheet Excel workbooks.

The software preserves traceability between operational requirements, objective evidence, representative hazards, quantitative risk assessment, readiness indicators, and critical-control findings. The Streamlit interface supports practical review of imported data and provides downloadable outputs for independent recalculation and audit. The source code, demonstration datasets, documentation, and technical review are maintained in a public GitHub repository, while a live deployment provides direct access to the interactive dashboard.

The software remains a proof-of-concept and computational verification platform rather than validated operational software. Automatically inferred values are labeled as estimated where applicable, synthetic records are explicitly identified, and analytical outputs must be interpreted in relation to data quality and operational context. Formal software validation, cybersecurity review, governance approval, and operational evaluation are required before institutional use.

2.8 External datasets for computational verification

External data sources were selected to evaluate whether the software could ingest, standardize, analyze, and export datasets with different structures. Their role is computational verification of the application rather than scientific validation of the MOBRA framework. No external dataset was used to establish universal readiness thresholds or to judge the safety of a country, institution, or laboratory.

Canadian laboratory incident data represent operational event reporting and follow-up. The source categories include Exposure Incident Notification Data, Exposure Incident Follow-up Data, Non-exposure Incident Data, and Other Event Data. These records are suitable for testing incident classification, exposure and root-cause analysis, corrective-action summaries, temporal trends, pathogen- or activity-based distributions, and risk-matrix generation. The principal limitation is that reported fields may not contain directly standardized MOBRA likelihood, consequence, or requirement scores; therefore, any software-derived mappings must be transparent and labeled as estimated.

The Global Health Security Index 2021 provides national-level indicators for 195 countries, including biosafety, biosecurity, laboratory systems, pathogen inventories, facility security, surveillance, preparedness, and emergency response. It is suitable for testing ranking tables, cross-country comparisons, scatter plots, correlation analysis, and exploratory mapping between external indicators and MOBRA assessment domains. These indicators describe national systems and should not be interpreted as direct measurements of individual mobile-laboratory readiness.

High-risk pathogen exposure-event records provide a hazard-centered structure that may include the pathogen, exposure type and mechanism, laboratory activity, containment level, number of exposed persons, root cause, outcome severity, and corrective actions. Such data are particularly useful for testing the Representative Hazard Register, likelihood and consequence estimation, risk-score calculation, risk categorization, and population of the 5 × 5 likelihood-consequence matrix.

Three synthetic datasets were also specified for controlled software testing: sample_canadian_incidents.csv, sample_ghs_index.csv, and sample_high_risk_exposures.csv. Each dataset contains diverse artificial records and is explicitly labeled as synthetic. Controlled synthetic data allow expected outputs, edge cases, missing values, inconsistent labels, and validation rules to be tested without representing real incidents or institutional performance.

Dataset	Analytical level	Primary MOBRA use	Key strengths	Main limitations
Canadian laboratory incident data	Operational incident and follow-up records	Incident classification, cause/exposure trends, corrective actions, risk analytics	Real-world event-reporting structure and multiple incident categories	Heterogeneous fields; no standardized MOBRA scores; reporting bias and missingness
GHS Index 2021	National preparedness and security indicators	Readiness comparisons, rankings, correlations, domain mapping	Broad international coverage and standardized indicators	National—not laboratory—level; contextual interpretation required
High-risk pathogen exposure events	Hazard and exposure-event records	Hazard register, L/C estimation, risk score and 5×5 matrix	Direct relevance to exposure mechanisms and consequences	Variable definitions, availability, and completeness across sources
Synthetic MOBRA datasets	Controlled artificial test records	Unit tests, edge cases, import/export and reproducibility checks	Known expected values; safe and repeatable testing	Not real-world evidence and cannot validate operational performance

Table 2A. Comparative role of external and synthetic datasets in computational verification of the MOBRA software. These datasets test software behavior and interoperability and do not constitute scientific or institutional validation of the framework.

2.9 Validation plan

Normal validation of the MOBRA framework is planned as a staged process. In the first phase, a multidisciplinary expert panel with relevant expertise in laboratory biosafety, biosecurity, biorisk management, mobile laboratory operations, quality management, and emergency response will evaluate the relevance, clarity, completeness, and operational feasibility of the initial Operational Requirements Library (ORL). Expert ratings will support item-level and scale-level content-validity analysis and inform refinement, retention, modification, or removal of individual requirements.

In the second phase, **inter-assessor reliability** will be evaluated by having two or more independent trained assessors apply the MOBRA instrument to the same operational setting under comparable conditions. Agreement between assessors will be examined to determine the reproducibility of requirement scoring and to identify items requiring clearer definitions, evidence criteria, or assessor guidance.

In the third phase, **multi-mission and multi-site operational pilot testing** will evaluate framework feasibility and usability across different mobile biological laboratory environments and deployment scenarios. Pilot evaluation will examine assessment completion time, availability and completeness of objective evidence, missing data, user acceptability, responsiveness to corrective and preventive actions (CAPA), and stability of readiness measurements across repeated assessments.

Institutional permissions and relevant ethics or governance review will be obtained where required. The results of these validation phases will inform refinement of the ORL, scoring rules, risk-acceptance criteria, potential weighting strategies, and operational decision thresholds before institutional implementation or broader adoption of the framework.

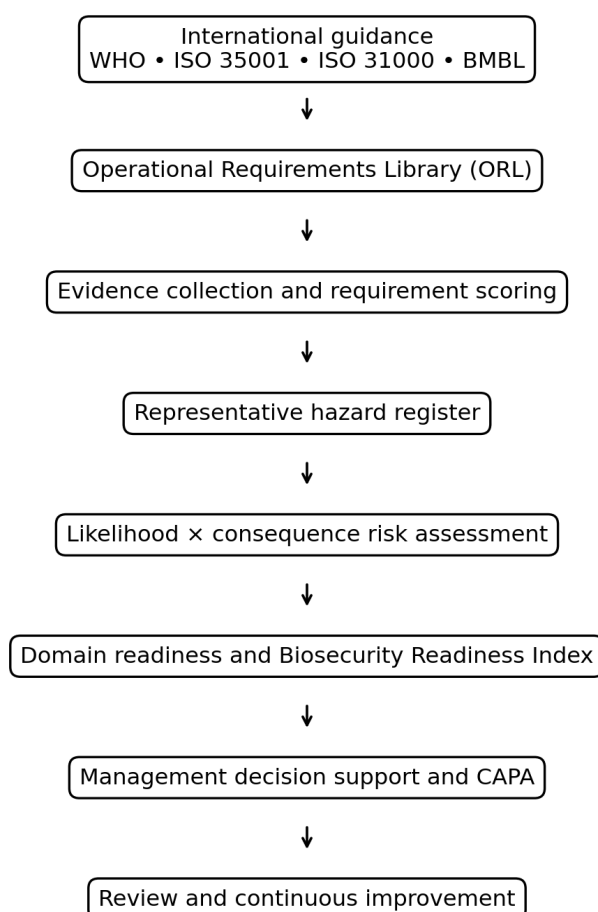


Figure 3. Overall MOBRA operational workflow. International guidance is translated into measurable operational requirements and objective evidence, followed by representative hazard assessment, quantitative risk classification, domain-level readiness measurement, calculation of the Biosecurity Readiness Index (BRI), and management decision support. Assessment findings inform corrective and preventive actions and continuous improvement.

3. Results

3.1 Framework outputs

Framework implementation generated four integrated operational outputs: **(i)** a structured Operational Requirements Library (ORL), **(ii)** an evidence-based assessment record, **(iii)** a representative hazard and risk-assessment module, and **(iv)** a readiness and management decision-support module. Operational requirements can be organized either by functional assessment domain or according to the operational lifecycle of a mobile biological laboratory. This dual representation supports both domain-oriented scientific auditing and mission-oriented field assessment. Collectively, these outputs establish a transparent and traceable pathway from international guidance to objective evidence, quantitative risk assessment, readiness measurement, and management decision support.

Figure 4 illustrates the dual operational representation adopted by MOBRA, allowing operational requirements to be interpreted either according to assessment domains or according to the chronological sequence of mission activities.



Operational requirements are mapped to mission stages and assessment domains

Figure 4. Mobile biological laboratory operational lifecycle supporting the MOBRA assessment process. Operational requirements can be assessed either by functional domain or according to the chronological sequence of activities performed before, during, and after deployment, supporting both domain-oriented auditing and mission-oriented field assessment.

3.2 Representative hazard register

The representative hazard register comprises **24 operational hazards** covering biological, engineering, laboratory operations, specimen security, physical security, personnel reliability, information security, transportation and logistics, waste management, and emergency-management concerns. The hazards were selected to provide representative coverage of the principal operational biosecurity risks that may arise during mobile biological laboratory deployment.

Each representative hazard is linked conceptually to one or more operational requirements and the associated objective evidence, supporting traceability between identified operational expectations, verification evidence, and risk prioritization. The register is intentionally representative rather than exhaustive and therefore requires institution-specific review and validation before operational implementation. The likelihood and consequence ratings presented in the initial register are prototype design values used to demonstrate framework functionality and should not be interpreted as observed risks or as an institutional risk assessment.

ID	Hazard	Category	L	C	Score	Level
H01	Aerosol generation during specimen opening	Biological	4	5	20	Extreme
H02	Loss of negative pressure	Engineering	4	5	20	Extreme
H03	Class III glovebox leak	Engineering	2	5	10	High
H04	Class II BSC airflow failure	Engineering	3	4	12	High
H05	HEPA filter integrity failure	Engineering	2	5	10	High
H06	Specimen misidentification	Laboratory operations	3	5	15	High
H07	Broken chain of custody	Specimen security	3	4	12	High
H08	Specimen leakage during transport	Transportation	3	5	15	High
H09	Cold-chain interruption	Logistics	3	3	9	Moderate
H10	Unauthorized laboratory access	Physical security	2	5	10	High
H11	Credential or key misuse	Physical security	2	4	8	Moderate
H12	Theft or diversion of specimens	Specimen security	2	5	10	High
H13	Unauthorized disclosure of sensitive data	Information security	3	4	12	High
H14	Loss of secure communications or connectivity	Information security	3	3	9	Moderate
H15	Unauthorized alteration of laboratory records	Information security	2	4	8	Moderate
H16	Competency gap for mission-critical tasks	Personnel reliability	3	4	12	High
H17	Fatigue-related procedural error	Personnel reliability	4	3	12	High
H18	Insider misuse of authorized access	Personnel reliability	2	5	10	High
H19	Surface or equipment decontamination failure	Laboratory operations	3	5	15	High
H20	Autoclave cycle or verification failure	Waste management	2	4	8	Moderate
H21	Incorrect segregation or containment of biohazardous waste	Waste management	3	4	12	High
H22	Emergency power or utility loss	Emergency management	3	4	12	High
H23	Fire, collision, or vehicle emergency during mission operations	Emergency management	2	5	10	High
H24	Delayed incident reporting, escalation, or corrective response	Emergency management	3	4	12	High

Table 3. Initial MOBRA representative hazard register comprising 24 operational hazards. L = likelihood; C = consequence; risk score = L × C. Likelihood and consequence ratings are prototype design values provided for framework demonstration and require context-specific validation before operational use.

3.3 Heat mapping and prototype analytics

The MOBRA prototype generated a **5 × 5 likelihood–consequence heat map** to visualize the distribution of representative hazards across the predefined risk matrix. Each matrix cell retains the color corresponding to its predefined risk category, while the numerical value displayed within the cell indicates the number of representative hazards assigned to the corresponding likelihood–consequence combination.

This visualization distinguishes **risk severity**, represented by the predefined cell color, from **hazard frequency**, represented by the numerical count within each cell. The heat map therefore provides a rapid visual summary of both the distribution and relative risk intensity of representative hazards without altering the underlying risk classification rules.

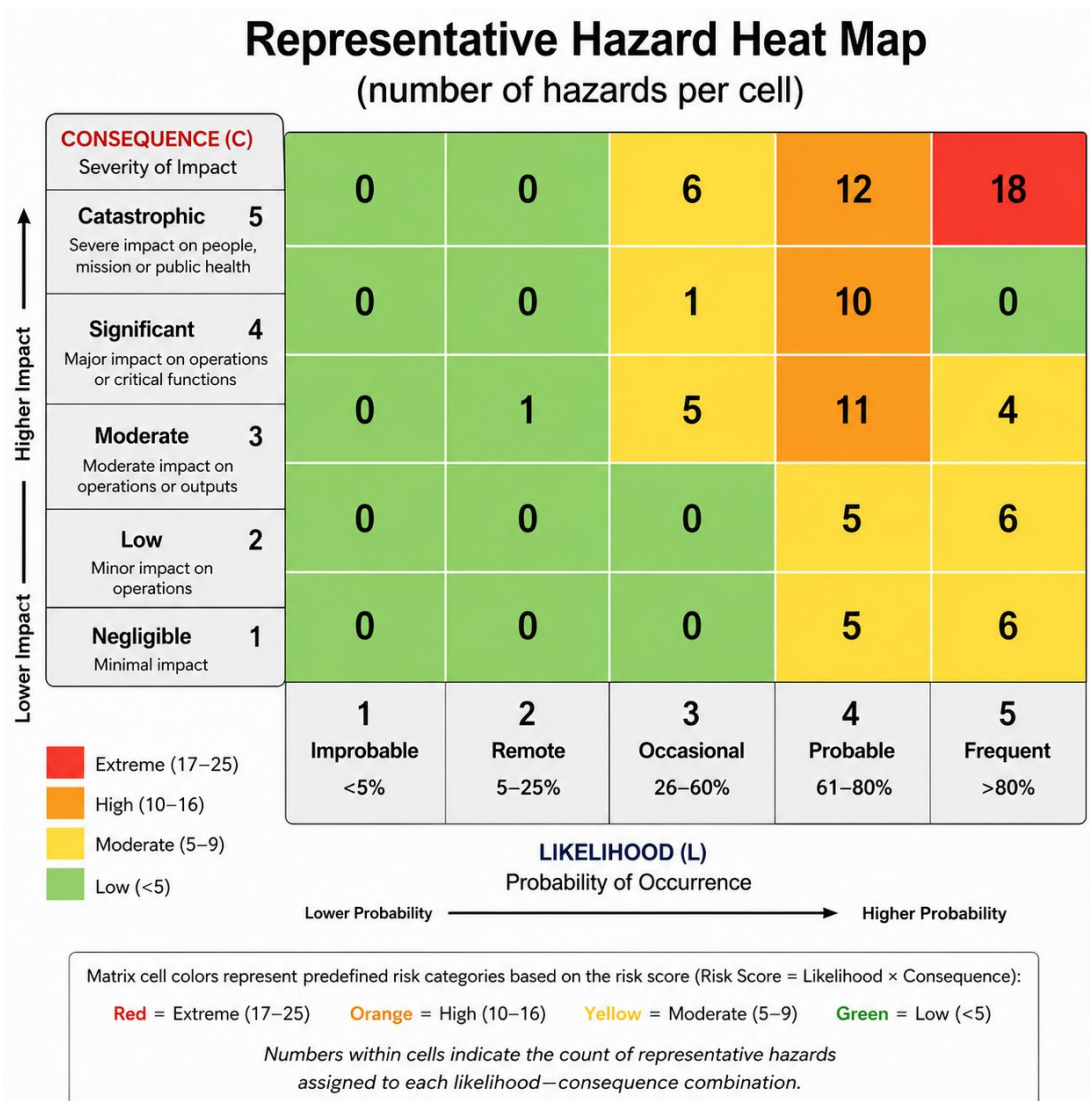


Figure 5. Representative hazard distribution heat map generated by the MOBRA prototype. Matrix cell colors represent predefined risk categories based on the calculated risk score (green = low, 1–4; yellow = moderate, 5–9; orange = high, 10–16; red = extreme, 17–25). Numbers within each cell indicate the number of representative hazards assigned to the corresponding likelihood–consequence combination. Cell colors represent risk severity and are independent of the number of hazards displayed within each cell.

3.4 Synthetic demonstration

A synthetic dataset was used to demonstrate the complete end-to-end functionality of the MOBRA framework and its supporting Python prototype. The demonstration dataset enabled execution of the full analytical workflow, including requirement-level scoring, domain-readiness calculation, representative hazard classification, risk visualization, calculation of the overall Biosecurity Readiness Index (BRI), and generation of interactive decision-support outputs.

The synthetic demonstration produced an illustrative overall **BRI of 81.0%**. This value, together with all requirement scores, domain-level readiness values, likelihood and consequence ratings, and prototype-generated outputs, was created solely to demonstrate framework functionality and computational reproducibility. These values **do not represent an assessment of the Jordanian reference mobile biological laboratory or of any other institution** and should not be interpreted as evidence of actual operational readiness or performance.

The demonstration confirmed the technical feasibility of implementing the proposed framework as a transparent and reproducible digital workflow capable of integrating operational requirements, objective evidence, representative hazards, quantitative risk assessment, and readiness indicators within a unified decision-support architecture. Formal expert validation, inter-assessor reliability testing, and multi-site operational evaluation remain necessary before institutional adoption or operational deployment.

Domain	Requirements	Synthetic readiness (%)
Emergency Incident Management	6	66.7
Engineering Controls	6	83.3
Information Security	6	83.3
Laboratory Operations	6	76.7
Mission Planning	6	83.3
Personnel Reliability	6	86.7
Physical Security	6	83.3
Specimen Security	6	93.3
Transportation Logistics	6	76.7
Waste Management Continuous Improvement	6	76.7

Table 4. Interactive MOBRA prototype dashboard generated using synthetic demonstration data. The dashboard integrates domain-level readiness indicators, representative hazard-risk visualization, and the overall Biosecurity Readiness Index (BRI). All displayed values are illustrative and do not represent an institutional assessment.

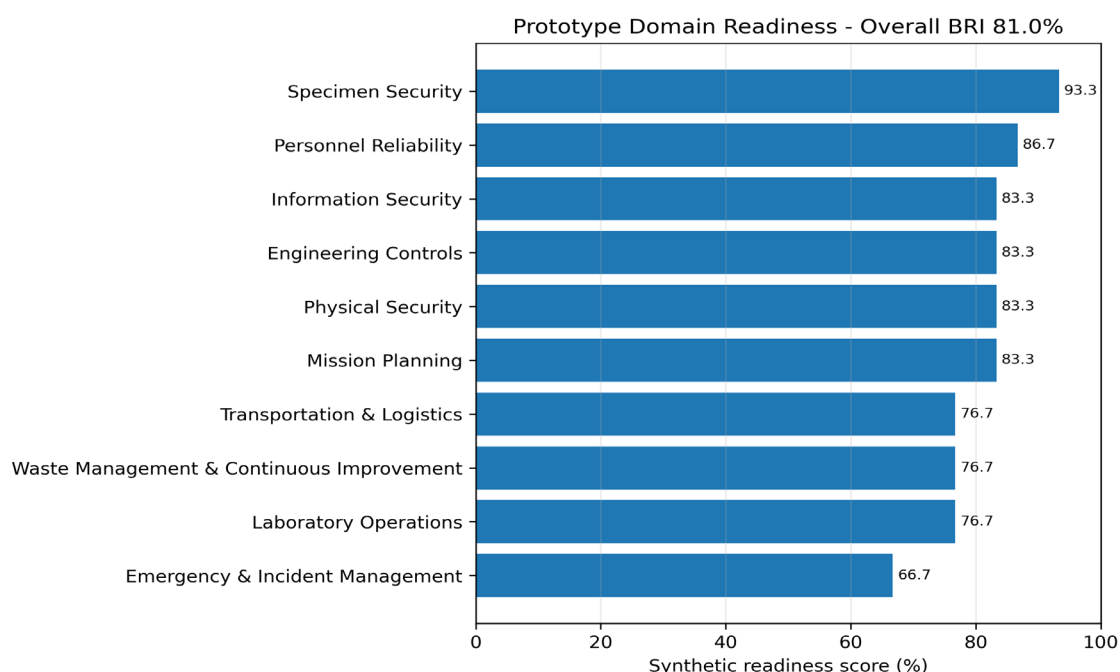


Figure 6. Synthetic domain-readiness output. Percentages equal observed domain scores divided by maximum achievable scores. All values are illustrative and do not represent institutional performance.

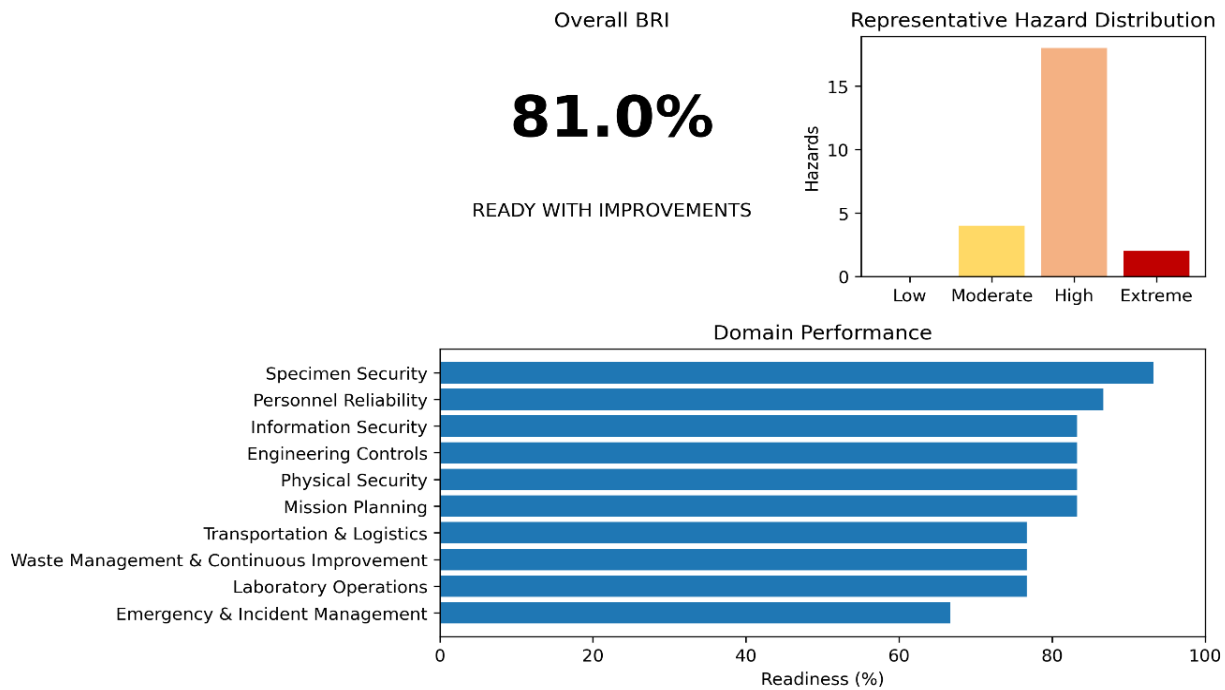


Figure 7. Integrated MOBRA prototype dashboard. The dashboard combines domain readiness, hazard distribution, risk categorization, and the overall Biosecurity Readiness Index. All displayed values originate from the synthetic demonstration dataset.

3.5 Live interactive dashboard and public implementation

The MOBRA software was deployed on Streamlit Community Cloud as a publicly accessible interactive dashboard. The deployment enables users to upload compatible datasets, inspect data quality, map source columns to MOBRA variables, calculate risk and readiness indicators, visualize results, and download analytical outputs. The public implementation demonstrates reproducibility and accessibility of the proof-of-concept workflow without implying operational certification or institutional endorsement. The live dashboard link, source-code repository, supporting documentation, and machine-readable QR code are provided in Appendix C (Appendix 3).

3.6 Output files (Decision-Support Logic)

MOBRA integrates aggregate readiness, domain-level performance, representative hazard assessment, and critical-control status within a unified decision-support process. The overall Biosecurity Readiness Index (BRI) provides a quantitative summary of assessed readiness, while domain-level indicators identify specific operational areas requiring improvement. Representative hazard analysis provides complementary risk information by identifying hazards requiring prioritization, mitigation, or escalation.

Deployment or operational decisions are therefore not based on the BRI alone. The framework requires consideration of the overall BRI together with domain-level readiness, the completeness and quality of objective evidence, residual risk status, and any identified failure of mission-critical containment or biosecurity controls. Under the proposed **critical-control override**, an extreme residual risk or failure of a mission-critical control may prevent deployment approval regardless of an otherwise high aggregate readiness score.

This integrated logic enables assessment findings to be translated into prioritized management actions, including corrective and preventive actions (CAPA), additional evidence collection, targeted training, control enhancement, reassessment, or escalation to institutional decision-makers. MOBRA therefore functions not only as a readiness measurement framework but also as a structured operational decision-support architecture linking assessment findings to risk-informed management action.

Assessment finding	Illustrative management response
High overall readiness with no critical-control failure or extreme residual risk	Deployment may be considered, subject to institutional authorization and mission-specific review.
Moderate readiness or identified domain-level weaknesses	Corrective actions and targeted control improvements should be implemented before deployment or within an institutionally approved timeframe.
Low readiness or substantial evidence gaps	Deployment should generally be deferred pending corrective actions, completion of required evidence, and reassessment.
Extreme residual risk	Deployment should not proceed until the identified risk has been reduced to an institutionally acceptable level.
Failure of a mission-critical containment or biosecurity control	The critical-control override is activated; deployment approval may be withheld regardless of the overall Biosecurity Readiness Index (BRI).

Table 5. Illustrative MOBRA decision-support rules for prototype demonstration

Note: These decision-support rules are illustrative prototype design rules and do not constitute validated deployment thresholds. Final decision criteria, readiness thresholds, risk-acceptance levels, and escalation procedures require institutional governance and formal validation.

Software output	Purpose
domain_summary.csv	Domain-level readiness and requirement counts
scored_hazard_register.csv	Likelihood, consequence, risk score, and category
hazard_heatmap.html	Interactive color-coded 5 × 5 matrix
domain_dashboard.html	Interactive domain-readiness visualization
summary.json	Machine-readable executive summary

Table 6. MOBRA Python prototype outputs. Open formats facilitate independent review, recalculation, reproducibility, and future software integration.

Prototype capabilities
• Operational requirement scoring and domain-readiness calculation.
• Automated risk scoring and categorical risk assignment.
• Color-coded 5 × 5 hazard matrix and interactive visual outputs.
• CSV, JSON, and HTML export for audit and independent recalculation.
• Modular architecture supporting future integration and extension.

4. Discussion

This study presents MOBRA as a requirements-based operational biosecurity readiness and decision-support framework designed specifically for the complex and dynamic operating environment of mobile biological laboratories. The principal contribution of MOBRA is not the introduction of new biosafety or biosecurity principles, but the systematic translation of established international guidance into a structured, measurable, evidence-based, and digitally implementable operational assessment architecture. By integrating an Operational Requirements Library (ORL), objective evidence, representative hazard assessment, quantitative readiness indicators, and management decision-support logic, the framework provides a traceable pathway from normative guidance to operational action.

Unlike conventional checklist-based approaches that primarily document the presence or absence of individual controls, MOBRA is designed to evaluate the maturity and evidentiary strength of operational implementation. The six-level assessment scale enables differentiation between absent, informal, partially implemented, implemented, objectively verified, and continuously reviewed controls. This distinction is particularly relevant in mobile laboratory environments, where the nominal existence of a procedure or control does not necessarily demonstrate its effective implementation under changing field conditions.

The integration of readiness assessment with representative hazard analysis further distinguishes MOBRA from purely compliance-oriented approaches. The Biosecurity Readiness Index (BRI) and domain-level readiness indicators provide quantitative summaries of assessment findings, while the hazard register and critical-control override preserve the

importance of risk-specific findings that may not be adequately represented by an aggregate score. Consequently, a high overall BRI cannot compensate for an extreme residual risk or the failure of a mission-critical containment or biosecurity control. This approach positions MOBRA as a risk-informed decision-support framework rather than a simple scoring or ranking instrument.

4.1 Operational Requirements Library and Traceability

A central feature of MOBRA is the Operational Requirements Library (ORL), which functions as a structured operational knowledge base rather than as a static checklist. The initial library organizes 60 measurable operational requirements across ten assessment domains, providing a systematic mechanism for translating broad international biosafety, biosecurity, and biorisk-management principles into assessable operational expectations. This structure is intended to support consistent assessment while maintaining flexibility for future refinement through expert validation and field testing.

The value of the ORL lies particularly in its emphasis on traceability. Operational requirements are connected to expected objective evidence and, where applicable, to representative hazards, enabling assessors and decision-makers to examine not only whether a requirement has been addressed but also how its implementation is demonstrated and what operational risks may be associated with inadequate control. This traceability supports a transition from document-centered compliance toward evidence-informed operational assessment.

The ORL should nevertheless be considered an initial framework component rather than a finalized or universally applicable requirements catalogue. The current allocation of six requirements to each of the ten domains was adopted as a transparent prototype design choice and does not imply that all domains necessarily carry equal operational importance. Expert content validation and operational testing may identify requirements that should be modified, removed, added, or assigned different levels of criticality. Future development may therefore incorporate expert-derived weighting or critical

4.2 Risk-Informed Readiness and the Role of the BRI

The Biosecurity Readiness Index (BRI) was developed to provide a concise quantitative summary of operational readiness across the MOBRA assessment framework. By aggregating requirement-level scores, the BRI can facilitate communication of overall readiness status and support comparison across repeated assessments. Domain-level readiness indicators complement the overall index by identifying specific operational areas in which corrective action, additional evidence, or targeted improvement may be required.

However, aggregate readiness scores may conceal critical weaknesses if interpreted in isolation. For this reason, MOBRA does not treat the BRI as a standalone deployment criterion. The framework combines aggregate readiness with domain-level performance, objective-evidence quality, representative hazard assessment, residual risk, and critical-control findings. The proposed critical-control override ensures that an extreme residual risk or failure of a mission-critical containment or biosecurity control can take precedence over an otherwise favorable aggregate readiness score.

This approach reflects the distinction between **readiness measurement** and **risk acceptance**. The BRI summarizes the assessed maturity of operational requirements, whereas decisions regarding whether a particular residual risk is acceptable remain dependent on institutional governance, mission context, risk appetite, and applicable regulatory or organizational requirements. Accordingly, the prototype risk categories and decision-support rules presented in this study should be interpreted as provisional framework design parameters rather than validated universal thresholds.

The illustrative BRI of **81.0%** generated in the synthetic demonstration should therefore be interpreted solely as evidence that the computational workflow can aggregate requirement-level assessments into a reproducible readiness indicator. It does not establish an acceptable deployment threshold, nor does it represent the readiness of the Jordanian reference laboratory or any other institution. Future validation should determine whether the unweighted BRI provides sufficient operational discrimination and whether expert-derived weighting, criticality-based scoring, or alternative aggregation approaches improve its validity and decision-support value.

4.3 Digital Implementation, Transparency, and Reproducibility

The Python prototype demonstrates the technical feasibility of implementing MOBRA as a transparent and reproducible digital assessment workflow. By automating input validation, requirement-level scoring, domain-readiness calculations, hazard-risk classification, BRI calculation, and visual analytics, the prototype reduces the need for manual calculation and provides a consistent computational pathway from assessment data to decision-support outputs.

The use of open and machine-readable output formats, including CSV, JSON, and HTML, supports independent review, recalculation, auditability, and reproducibility. Interactive visualizations, including the likelihood–consequence heat map, domain-readiness summaries, and integrated dashboard, can further assist assessors and decision-makers in identifying patterns, operational weaknesses, and priority areas requiring management attention. Importantly, these visual outputs are intended to complement rather than replace examination of the underlying requirement-level evidence and risk information.

The prototype also demonstrates the potential for future integration with laboratory information management systems (LIMS), automated reporting platforms, and other institutional digital infrastructures. Such integration could enable more efficient evidence management, longitudinal readiness monitoring, and structured follow-up of corrective and preventive actions (CAPA). However, these capabilities remain prospective and were not evaluated in the present study.

The current prototype should therefore be regarded as a **proof-of-concept implementation rather than validated operational software**. Before institutional deployment, further development should address software validation, data governance, cybersecurity, user-access controls, version management, audit trails, and testing across different operational environments. These considerations are particularly important in biosecurity applications, where digital systems may process sensitive operational, specimen-related, or security-relevant information.

4.4 External-data interoperability and interpretive safeguards

Testing with operational incident records, national preparedness indicators, pathogen exposure-event structures, and controlled synthetic datasets demonstrates an important architectural feature of MOBRA: the software is not dependent on a single fixed input schema. Column mapping, data-quality checks, and modular analytical components allow heterogeneous sources to be transformed into a transparent common workflow while preserving the distinction between observed, mapped, and estimated values.

The selected sources are complementary rather than interchangeable. Canadian incident records reflect reported operational events and follow-up actions; GHS Index indicators describe national preparedness environments; high-risk exposure-event data emphasize hazard mechanisms and outcomes; and synthetic datasets provide controlled expected values for software testing. Their different units of analysis mean that outputs cannot be pooled or compared without explicit harmonization and contextual interpretation.

Most importantly, successful processing of these datasets represents computational verification only. It does not establish content validity, criterion validity, predictive validity, or institutional readiness. Data completeness, reporting practices, variable definitions, and selection bias can materially affect results. Consequently, automatically generated likelihood or consequence values must be labeled as estimated, and national indicators must not be used as proxies for the readiness of an individual laboratory without additional evidence.

4.5 Applicability and Adaptability Across Mobile Laboratory Missions

MOBRA was designed as a mission-adaptable framework rather than as an assessment instrument restricted to a specific pathogen, specimen type, or diagnostic methodology. Although the initial framework was informed by the technical and operational characteristics of a truck-based mobile biological laboratory, its requirements-based architecture allows the assessment process to be adapted to different mobile laboratory configurations, deployment environments, and diagnostic missions.

This adaptability is particularly relevant because mobile biological laboratories may support a wide range of activities, including molecular and genetic testing, microbiological diagnostics, outbreak investigation, pathogen detection, genomic surveillance, and other specialized laboratory functions. Different missions may involve distinct biological materials, analytical platforms, containment requirements, transportation conditions, information-security considerations, and waste-management processes. The ORL can therefore provide a common operational assessment structure while allowing individual requirements, objective-evidence criteria, and representative hazards to be refined according to mission-specific needs.

The framework should not, however, be interpreted as implying that identical controls or risk profiles apply to all specimens or diagnostic activities. Application of MOBRA to a new mission should be accompanied by activity-specific and context-specific risk assessment, review of applicable regulatory and institutional requirements, and appropriate modification of the ORL and hazard register. This modular approach may allow MOBRA to retain a consistent readiness-assessment

architecture while accommodating the operational diversity of molecular, genetic, microbiological, and other biological laboratory activities.

Future multi-site and multi-mission validation will be necessary to determine the extent to which the initial MOBRA structure can be generalized across different mobile biological laboratory platforms and operational contexts. Such evaluation may also identify a set of core requirements applicable across most missions together with mission-specific modules tailored to particular diagnostic technologies, specimen types, or deployment scenarios.

4.6 Limitations and Validation Priorities

Several limitations should be considered when interpreting the present study. First, MOBRA is currently a framework-development and proof-of-concept study. The initial 60-item Operational Requirements Library (ORL) has not yet undergone formal expert content validation, and the equal allocation of six requirements across each of the ten assessment domains represents a prototype design decision rather than an empirically validated structure. Expert review may therefore result in the modification, addition, removal, reclassification, or reprioritization of individual requirements.

Second, the 24-item hazard register is representative rather than exhaustive, and the likelihood and consequence ratings used in the prototype are design values intended to demonstrate framework functionality. They should not be interpreted as observed risk estimates or institutional risk assessments. Similarly, the proposed risk categories, critical-control override logic, and illustrative decision-support rules require context-specific validation and alignment with institutional governance, risk appetite, and applicable regulatory requirements before operational use.

Third, the readiness results presented in this study were generated using synthetic demonstration data, while the external datasets were selected for computational verification of import, mapping, cleaning, visualization, and analytical behavior. Neither the illustrative BRI of 81.0% nor results derived from incident, exposure, or national indicator data represent institutional readiness measurements. External sources may contain missing values, reporting bias, inconsistent definitions, and different units of analysis; any inferred likelihood, consequence, or MOBRA-domain mapping must therefore be explicitly documented and labeled as estimated.

Fourth, the initial BRI uses equal weighting of operational requirements. Although this approach improves transparency and ease of interpretation during prototype development, it may not adequately reflect differences in operational criticality among requirements or domains. Future validation should examine whether expert-derived weighting, criticality-based classifications, or alternative aggregation methods improve the validity, sensitivity, and operational usefulness of readiness measurement without reducing interpretability.

Finally, framework development was informed by a single operational reference environment, which may limit immediate generalizability to other mobile laboratory configurations and deployment contexts. Formal validation should therefore include multidisciplinary expert content review, inter-assessor reliability testing, and multi-site and multi-mission operational pilots. These studies should evaluate feasibility, scoring consistency, evidence availability, assessment completion time, responsiveness to corrective actions, and stability across repeated assessments. Validation findings should guide refinement of the ORL, hazard register, scoring criteria, risk-acceptance rules, potential weighting strategies, and decision-support thresholds before broader institutional adoption.

4.7 Future Development and Research Directions

Future development of MOBRA should focus on systematic validation, refinement, and evaluation across diverse operational settings. The first priority is formal content validation of the Operational Requirements Library (ORL) by a multidisciplinary expert panel representing laboratory biosafety, biosecurity, biorisk management, mobile laboratory operations, quality management, emergency response, and relevant technical disciplines. This process should evaluate the relevance, clarity, completeness, and operational feasibility of individual requirements and determine whether additional requirements, domain restructuring, or criticality classifications are necessary.

Subsequent research should evaluate inter-assessor reliability and the reproducibility of MOBRA scoring when applied by independently trained assessors. Clearer evidence criteria and standardized assessor guidance may be required for requirements demonstrating lower levels of agreement. Multi-site and multi-mission pilot studies should then examine the framework under different mobile laboratory configurations, deployment conditions, specimen types, and diagnostic activities to assess feasibility, usability, adaptability, and generalizability.

Further methodological development should also investigate alternative approaches to readiness aggregation. Expert-derived weighting, requirement criticality, domain-specific weighting, and sensitivity analysis may help determine whether the initial unweighted BRI can be improved while preserving transparency and interpretability. Longitudinal studies could

additionally evaluate whether changes in MOBRA readiness indicators correspond to corrective and preventive actions (CAPA), training interventions, operational changes, or repeated deployment experience.

From a digital perspective, future versions of the prototype could incorporate structured evidence repositories, automated CAPA tracking, longitudinal readiness monitoring, role-based access controls, enhanced audit trails, and integration with laboratory information management systems (LIMS) or institutional reporting platforms. Any such development should be accompanied by appropriate software validation, cybersecurity controls, data-governance requirements, and institutional authorization.

Finally, future research may explore the development of a **core MOBRA requirements set** applicable across mobile biological laboratory missions, complemented by **mission-specific modules** for particular diagnostic technologies, specimen categories, biological risks, or deployment environments. This modular structure could improve adaptability while preserving a common and traceable operational readiness architecture. However, broader adoption should remain dependent on formal validation and evidence demonstrating that the framework is reliable, practical, and operationally useful across different contexts.

5. Conclusion

This study developed MOBRA as a requirements-based operational biosecurity readiness and decision-support framework for mobile biological laboratories. The framework integrates an Operational Requirements Library (ORL), evidence-based maturity scoring, representative hazard assessment, quantitative domain-readiness indicators, an overall Biosecurity Readiness Index (BRI), and critical-control decision logic within a single transparent and traceable operational architecture.

The supporting Python and Streamlit implementation demonstrated the technical feasibility of the complete MOBRA workflow, including structured data import, quality assessment, column mapping, automated requirement and hazard scoring, risk-matrix visualization, BRI calculation, critical-control logic, interactive dashboards, and reproducible exports. Testing pathways were defined for Canadian laboratory incident records, GHS Index 2021 indicators, high-risk pathogen exposure-event structures, and controlled synthetic datasets. These sources support computational verification only and do not replace formal scientific or operational validation.

MOBRA is designed to support the translation of established international biosafety, biosecurity, and biorisk-management guidance into measurable operational requirements and evidence-informed management action. Its modular architecture may also enable adaptation across different mobile laboratory configurations, diagnostic missions, specimen types, and biological laboratory activities, provided that appropriate mission-specific risk assessment and contextual adaptation are undertaken.

Formal expert content validation, inter-assessor reliability testing, software validation, and multi-site and multi-mission operational evaluation are required before institutional adoption. Subject to successful validation and refinement, MOBRA may provide a structured approach for strengthening operational biosecurity readiness, risk-informed decision-making, accountability, and continuous improvement in mobile biological laboratory operations.

Appendix A presents an independent validation study conducted before and after the implementation of the MOBRA framework. The study evaluated changes in relevant operational and performance indicators and demonstrated measurable improvement following framework implementation. Additional details regarding the study methodology, dataset characteristics, comparative pre- and post-implementation results, and performance outcomes are provided in Appendix A.

Declarations

Ethics: No patient-identifiable information or human-participant intervention was used in the framework and software prototype.

Data and code availability: The Python implementation of the MOBRA framework, including the interactive dashboard, source code, demonstration datasets, and supporting documentation, is publicly available. Source code: <https://github.com/modiabat-coder/mobra-python-dashboard> Interactive dashboard: <https://mobra-biosecurity-lab.streamlit.app/>

Software status and use limitation: The public application is a proof-of-concept research implementation for computational demonstration and technical testing. It is not a validated operational authorization system, and outputs must be interpreted with source-data quality, context, governance, and framework limitations.

Conflict of interest: The author declares no competing interests.

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Author contribution: Mohammad Ahmad Yousef E'Diabat conceived the project, supplied the operational case context, developed the framework, and prepared the manuscript and prototype materials.

framework applicability

The MOBRA framework is intentionally platform-independent and assay-agnostic. Rather than being limited to a specific pathogen, specimen type, or diagnostic method, it evaluates operational readiness and biorisk management using standardized assessment criteria. Consequently, the framework can be tailored to laboratories performing molecular diagnostics, microbial testing, genomic sequencing, pathogen surveillance, or other laboratory workflows without modification of its core methodology.

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Appendix A

Clinical Validation of the MOBRA Framework

A.1 Study Design and Reference Standard

A retrospective paired validation study was conducted using 320 nasopharyngeal swab specimens collected during routine SARS-CoV-2 molecular diagnostic testing. Molecular analysis was performed using the QuantStudio™ 5 Real-Time PCR System with a fluorescent-probe SARS-CoV-2 nucleic-acid detection assay. Each specimen was interpreted using two workflows: conventional laboratory interpretation (before MOBRA) and standardized MOBRA-guided interpretation (after MOBRA). Both interpretations were compared with a composite reference standard incorporating clinical assessment, chest imaging, SARS-CoV-2 serology, and confirmatory testing of a randomly selected subset of specimens by an independent laboratory.

Table A1. Validation dataset characteristics

Characteristic	Value
Study design	Retrospective paired validation
Number of specimens	320
Sample type	Nasopharyngeal swabs
Molecular platform	QuantStudio™ 5 Gen Real-Time PCR
Comparison	Before MOBRA vs after MOBRA Rusette
Reference standard	Composite clinical, radiological, serological, and external-laboratory evidence

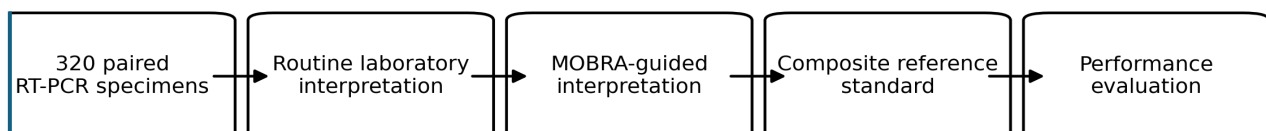
A.2 Standardized Result Classification

Original laboratory labels were standardized into four diagnostic categories to enable paired comparison and consistent statistical analysis.

Table A2. Standardized interpretation categories

Category	Operational definition
Negative	No SARS-CoV-2 target genes were detected with successful amplification of the human internal control.
Pending (Inconclusive)	Only one viral target detected (specific regions of the virus's RNA genome, the N (Nucleocapsid), ORF1ab (replicase complex), E (Envelope), and S (Spike) genes) with a valid human gene as an internal control; repeat or confirmatory testing required.
Positive	Two or more SARS-CoV-2 target genes detected.
Invalid	Internal-control failure or an inadequate specimen requiring repeat testing or recollection.

A.3 Validation Workflow



Overall diagnostic accuracy after MOBRA-guided interpretation: 89%

Figure A1. Paired operational validation workflow used to compare routine interpretation, MOBRA-guided interpretation, and the composite reference standard.

A.4 Results and Interpretation

Analysis of the 320 paired RT-PCR interpretations showed improved reporting consistency after application of the MOBRA-guided workflow. Compared with the composite reference standard, the standardized interpretation achieved an overall diagnostic accuracy of 89%. The workflow also reduced ambiguity associated with pending results and provided a clearer pathway for repeat or confirmatory testing. These findings support MOBRA as an operational decision-support approach for standardizing molecular-result interpretation; however, detailed class-specific estimates such as sensitivity and specificity require complete verified reference outcomes for all categories.

Table A3. Validation performance summary

Parameter	Result
Paired specimens evaluated	320
Reference approach	Composite reference standard
Overall diagnostic accuracy	89%
Reporting consistency	Improved
Ambiguous/pending interpretations	Reduced

A.5 Key Findings

- Validation used 320 paired routine clinical RT-PCR specimens.
- The same specimens were compared before and after MOBRA-guided interpretation.
- Final evaluation used a composite reference standard.
- MOBRA-guided interpretation achieved 89% overall diagnostic accuracy.
- Standardization improved reporting consistency and reduced ambiguity in pending results.

Appendix B

List of Abbreviations

Abbreviation	Full Term
JSON	JavaScript Object Notation
LIMS	Laboratory Information Management System
MOBRA	Mobile Biological Laboratory Operational Biosecurity Readiness Assessment
ORF1ab	Open Reading Frame 1ab
ORL	Operational Requirements Library
PCR	Polymerase Chain Reaction
qRT-PCR	Quantitative Reverse Transcription Polymerase Chain Reaction
RNA	Ribonucleic Acid
SARS-CoV-2	Severe Acute Respiratory Syndrome Coronavirus 2
WHO	World Health Organization
COVID-19	Coronavirus disease 2019

Abbreviation	Full Term
BMBL	Biosafety in Microbiological and Biomedical Laboratories
BRI	Biosecurity Readiness Index
BSC	Biological Safety Cabinet
CAPA	Corrective and Preventive Actions
CSV	Comma-Separated Values
DSR	Design Science Research
ELISA	Enzyme-Linked Immunosorbent Assay
HEPA	High-Efficiency Particulate Air
HTML	Hypertext Markup Language
ISO	International Organization for Standardization
JAF-RMS	Jordan Armed Forces–Royal Medical Services

Appendix C

Public Dashboard and Software Access



Figure C1. Machine-readable access card for the publicly deployed MOBRA Biosecurity Lab dashboard.

Interactive dashboard: <https://mobra-biosecurity-lab.streamlit.app/>

Source code and documentation: <https://github.com/modiabat-coder/mobra-python-dashboard>

The dashboard, code repository, and demonstration datasets are provided to support transparency, reproducibility, education, and computational verification. They do not constitute formal scientific validation of MOBRA or certification of any laboratory, institution, or country.